Q1: Write a MATLAB script to evaluate the function f(x) for x values range between -7 to 7 with increment of 0.05 then calculate the value of x which produces the largest value for f(x)

$$f(x) = \begin{cases} (x-5)^2 & if\ x > 0 \\ -(x-3)^2 + 34 & if\ x \le 0 \end{cases}$$

```
x = -7:0.05:7;
f1 = (x(x>0)-5).^2;
f2 = 34 - ((x(x<=0)-3).^2);

m = max(max(f1),max(f2));

disp(['The maximum value for the function is: '...
    num2str(m) ' with a value of x = ' num2str(x([f2,f1]==m))])
```

```
The maximum value for the function is: 25 with a value of x = 0
```

Q2: Solve the following differential equation:

$$(3+2y)\frac{dy}{dx} = 2\cos(2x), y(0) = -1$$

```
syms y(x)
eqn = diff(y,x) == (2*cos(2*x))/(3+2*y);
cond = y(0) == -1;
ysol(x) = dsolve(eqn,cond)
```

```
ysol(x) =
```

$$\frac{\sqrt{8\cos(x)\sin(x)+1}}{2} - \frac{3}{2}$$

Q3: Create the following matrices (MTX1 and MTX2):

$$MTX1 = \begin{bmatrix} 3 & 4 & 6 \\ -1 & 5 & 5 \\ 3 & -3 & 3 \end{bmatrix},\ MTX2 = \begin{bmatrix} 7 & 3 & 8 \\ 3 & 6 & 4 \\ -7 & 5 & 3 \end{bmatrix}$$

```
MTX1 = [3 4 6;
       -1 5 5;
        3 -3 3];
MTX2 = [7 3 8;
        3 6 4;
        -7 5 3];
```

1. Write a code to find maximum and minimum numbers among MTX1 and MTX2.

```
maxi = max(max(max(MTX1)),max(max(MTX2)))
```

```
maxi = 8
```

```
mini = min(min(min(MTX1)),min(min(MTX2)))
```

```
mini = -7
```

2. Write a code to exchange colum#2 of MTX1 with colum#3 of MTX2.

```
MTX1COPY = MTX1; MTX2COPY = MTX2;
[MTX1(:,2),MTX2(:,3)] = deal(MTX2(:,3),MTX1(:,2))
```

```
MTX1 = 3×3
     3     8     6
    -1     4     5
     3     3     3
MTX2 = 3×3
     7     3     4
     3     6     5
    -7     5    -3
```

3. Write a code to exchange row#1 of MTX1 with row#3 of MTX2.

```
[MTX1COPY(1,:),MTX2COPY(3,:)] = deal(MTX2COPY(3,:),MTX1COPY(1,:))
```

```
MTX1COPY = 3×3
    -7     5     3
    -1     5     5
     3    -3     3
MTX2COPY = 3×3
     7     3     8
     3     6     4
     3     4     6
```